

Deploy vibration and temperature sensors on draft gear mechanisms and journal bearings, connected to an onboard processing unit that analyzes data patterns indicative of wear or overheating

Executive Technology Brief

Domain: thermal

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'rolling-stock mechanical/electrical equipment failure' (FRA cause family E): 3 of 25 FRA Form 54 accidents in the retrieved window stem from equipment causes. Causes observed: 'Draft gear/mechanism broken or defective (including yoke)'; 'Journal (roller bearing) failure from overheating'; 'Other coupler and draft system defects, (CAR) (Provide detailed description in narrative)'; speeds 7–37 mph. LIMITS STAMPED INLINE: carrier-reported; cause codes are classifications, not proven root causes; above-threshold events only (Art. XXI.5 analog).

Mechanism: sensor-based condition monitoring

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: Deploy vibration and temperature sensors on draft gear mechanisms and journal bearings, connected to an onboard processing unit that analyzes data patterns indicative of wear or overheating

Governing Model: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: APPLICABLE; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: dT.

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: CONDITIONAL; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: NOT ESTABLISHED.

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k; one-dimensional gradient (lateral gradients neglected). Engaged variables: dT/dx.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.